

KMAT Cost Model

Brief Summary Documentation: CSS ML and FMIS ML

Phase 8 synthetic-data development and shadow evaluation summary

Overall status

CSS V2 and FMIS V2 are accepted as shadow candidates. Both remain advisory only because they were trained and evaluated on synthetic data. Neither model is allowed to change official cost, pricing, or quotation decisions.

Executive summary

Model	Purpose	Final shadow result	Cost authority
CSS V2	Predict high-scrap production batches.	Accepted for shadow. Balanced accuracy 76.24%; recall 86.49%.	Not allowed
FMIS V2	Forecast future material index and derive native FMIS multiplier.	Accepted for shadow with monitoring. Test MAE improved 36.11% versus naive.	Not allowed

Common governance rule

The deterministic KMAT cost engine remains the official financial authority. ML outputs are stored separately, versioned, and marked with **DEPLOYMENT_MODE = SHADOW**, **OFFICIAL_COST_IMPACT_ALLOWED_FLAG = FALSE**, and **BUSINESS_DECISION_ALLOWED_FLAG = FALSE**.

Core model locations in Snowflake

Model	Physical Snowflake ML object	Governance / output objects
CSS	KMAT_COST_MODEL_DB.CORE_ML.CSS_CLASSIFIER_V2	CSS_TEST_EVALUATION_V2, CSS_TEST_SHADOW_OUTPUT_V2, MODEL_REGISTRY
FMIS	KMAT_COST_MODEL_DB.CORE_ML.FMIS_FORECAST_V2	FMIS_FINAL_TEST_EVALUATION_V2, FMIS_FINAL_SHADOW_OUTPUT_V2, MODEL_REGISTRY

1. CSS ML - Scrap Risk Classification

Objective: Estimate the probability that a production batch will experience high scrap before the batch is completed. The output is a probability from 0 to 1 and a display score from 0 to 100.

0.12	86.49%	65.99%	76.24%
Selected probability threshold	Test recall	Test specificity	Balanced accuracy

Dataset and model design

- Synthetic CSS V2 dataset: 18,000 production batches from 2022-01 through 2025-12.
- High-scrap class rate: 18.794%.
- Chronological split: TRAIN through 2024-12, VALIDATION 2025-01 to 2025-06, TEST 2025-07 to 2025-12.
- Snowflake ML object: CSS_CLASSIFIER_V2.
- Threshold policy: recall at least 70% and specificity at least 60%, then maximize F2 on validation data.

Leakage control

The model uses only information available before or at production start. Outcome columns such as **ACTUAL_SCRAP_RATE_PCT**, **ACTUAL_DEFECT_COUNT**, and **ACTUAL_REWORK_COUNT** are excluded from model inputs.

Important features

Rank group	Representative features
Production history	Prior configuration scrap rate, prior line scrap rate, prior rework rate
Operating condition	Humidity, temperature, line utilization, recent downtime, machine health
Configuration risk	Configuration novelty, critical component count, engine/cab/wheel configuration
Supplier and workforce	Supplier quality, supplier defect rate, operator experience, automation level

Test result

Metric	Result	Interpretation
Precision	35.70%	About 36% of high-risk alerts were true high-scrap batches.
Recall	86.49%	Most high-scrap batches were detected.
Specificity	65.99%	About two-thirds of normal batches were correctly cleared.
False-positive rate	34.01%	Too high for automatic cost adjustment; acceptable for shadow review.
Accuracy	69.66%	Overall classification accuracy.
F2 score	67.33%	Recall-focused threshold performance.

CSS final decision

ACCEPTED_FOR_SHADOW. The model is suitable for risk-screening experiments and row-level audit analysis. It must not automatically change scrap rate, material cost, risk uplift, target price, or quotation values.

2. FMIS ML - Material Index Forecasting

Objective: Forecast the future realized material index for each commodity-region-supplier series, then derive the native FMIS multiplier used later by governed business rules.

2.1006	1.645%	36.11%	98.61%
Test ML MAE	Test ML WAPE	MAE improvement vs naive	Prediction interval coverage

Dataset and model design

- Synthetic FMIS V2 dataset: 30 commodity-region-supplier series, 96 monthly observations per series, 2,880 rows total.
- Chronological split: TRAIN 2018-01 to 2023-12, VALIDATION 2024-01 to 2024-12, TEST 2025-01 to 2025-12.
- Snowflake ML object: FMIS_FORECAST_V2.
- Exogenous inputs include approved energy, freight and FX forecasts, market PMI, port congestion, supplier lead time, supplier reliability, and contract escalation.
- The final test period was untouched until the model and validation process were completed.

FMIS technical output

FMIS_MULTIPLIER = FORECAST_MATERIAL_INDEX / CURRENT_APPROVED_INDEX

FMIS is not normalized into an artificial 0-100 financial input. The test multiplier range was **0.9674 to 1.0732**.

Test performance versus naive baseline

Metric	Raw ML	Naive baseline	Result
MAE	2.1006	3.2878	ML better
WAPE	1.6451%	2.5749%	ML better
MAPE	1.6583%	-	Low test error
Prediction interval coverage	98.61%	-	Strong test coverage
Invalid multipliers	0	-	Passed

Fallback experiment

A validation-derived per-series fallback policy was tested. It selected ML for 22 series and a naive fallback for 8 series. On the untouched test set, the hybrid MAE was **2.2965**, which was approximately **9.33% worse** than raw ML. The fallback policy was therefore rejected and preserved only for audit.

FMIS final decision

ACCEPTED_FOR_SHADOW_WITH_MONITORING. Raw ML is the selected shadow method. The model outperformed the naive baseline, generated no invalid multipliers, and remains prohibited from direct cost or business-decision use until validated on real historical data.

3. Governance, Storage, and Next Production Steps

Where the running models are stored

The trained models are Snowflake schema-level ML objects, not files in the internal stage and not ordinary database tables. Snowflake manages their internal artifacts. The models execute only when their prediction methods are called.

Area	CSS	FMIS
Trained model object	CORE_ML.CSS_CLASSIFIER_V2	CORE_ML.FMIS_FORECAST_V2
Prediction method	!PREDICT(...)	!FORECAST(...)
Stored test output	CSS_TEST_SHADOW_OUTPUT_V2	FMIS_FINAL_SHADOW_OUTPUT_V2
Evaluation record	CSS_TEST_EVALUATION_V2	FMIS_FINAL_TEST_EVALUATION_V2
Governance metadata	MODEL_REGISTRY	MODEL_REGISTRY

Production approval conditions

- Replace synthetic training data with approved historical MES, procurement, supplier, quality, and cost data.
- Rebuild all point-in-time features using production timestamps and data lineage controls.
- Run new chronological validation and retain a completely untouched out-of-time test period.
- Approve business thresholds, multiplier limits, fallback rules, and financial impact policies through the relevant owners.
- Operate in shadow mode with monitoring for drift, data quality, calibration, interval coverage, underforecast risk, and operational workload.
- Promote gradually through candidate, shadow, advisory, controlled, and production stages.

Final governance statement

The current results demonstrate that the Snowflake ML architecture, data splitting, evaluation, model storage, audit outputs, and safety flags work correctly. They do not establish real-world production accuracy. The deterministic KMAT cost engine remains the official source of cost and price until real-data approval is completed.

Phase 8 status: CSS and FMIS shadow development complete; official cost authority remains disabled; production approval awaits real historical data.